

HTML

(Hypertext Markup Language)

What is HTML?

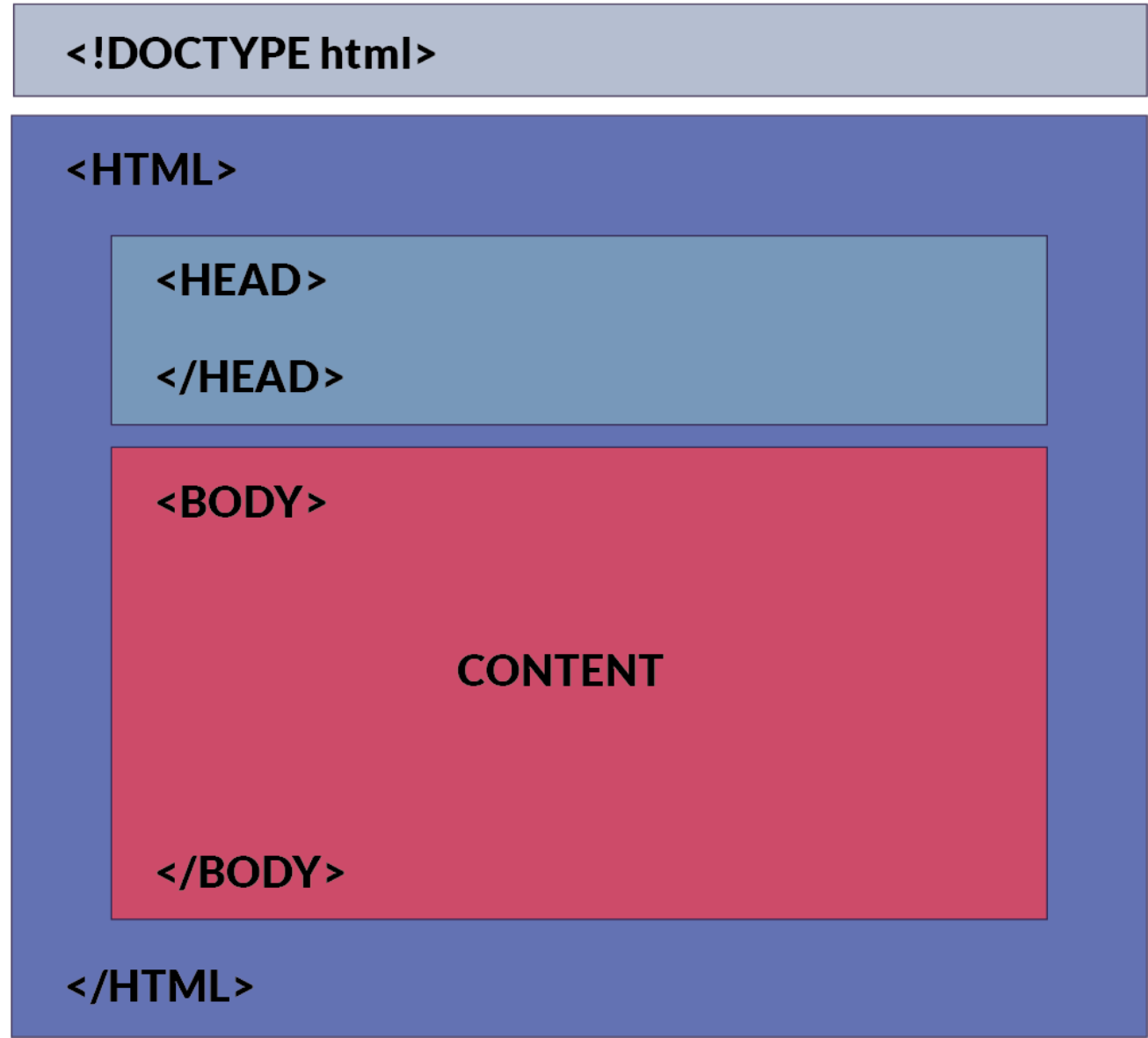
- HTML (Hypertext Markup Language) is the standard markup language for creating and structuring web pages.
- Purpose of HTML in Web Development:

HTML provides the foundation for building web pages by defining the structure and content of documents displayed in web browsers.

HTML Basics: HTML Structure

- An HTML document consists of a series of elements that define the structure and content of the web page.
- The document begins with a `<!DOCTYPE html>` declaration, followed by the `<html>`, `<head>`, and `<body>` elements.

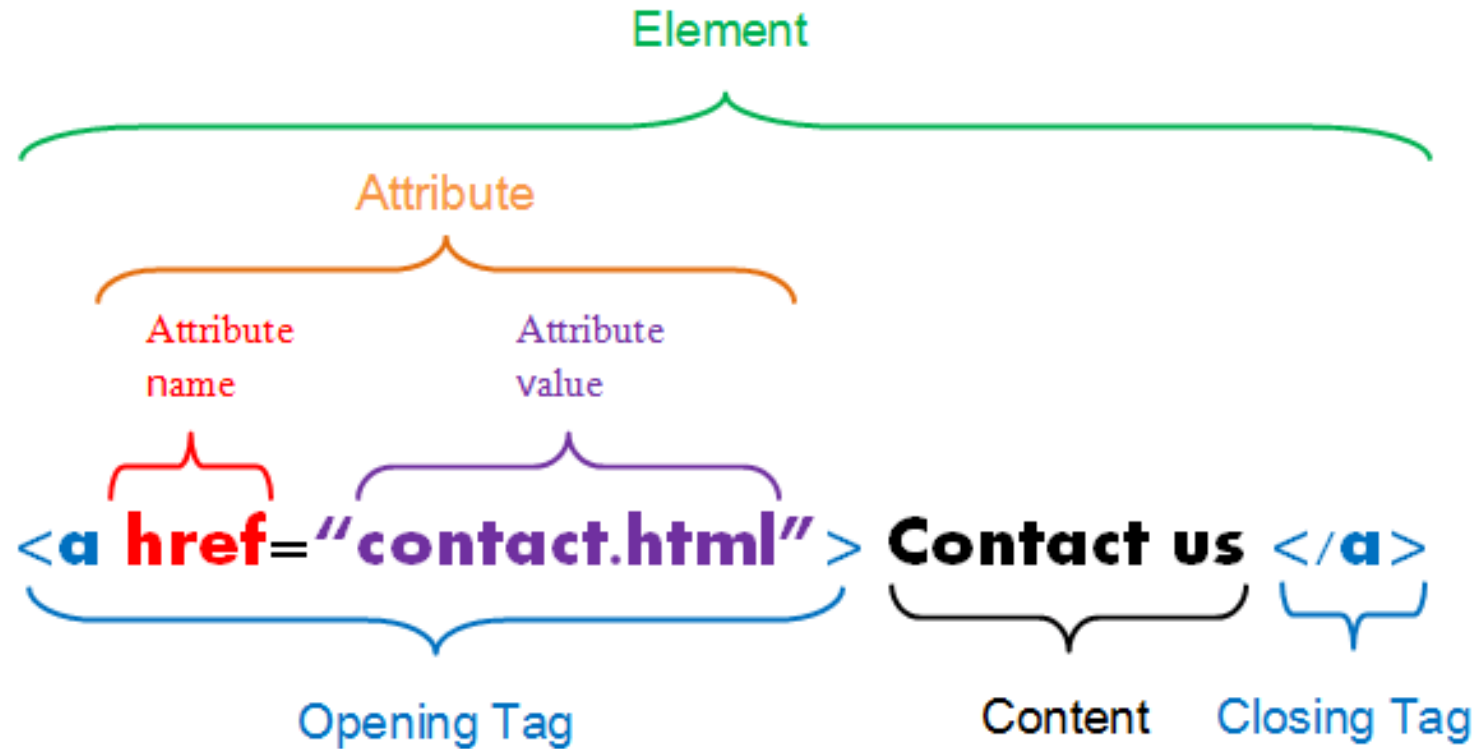
HTML Structure



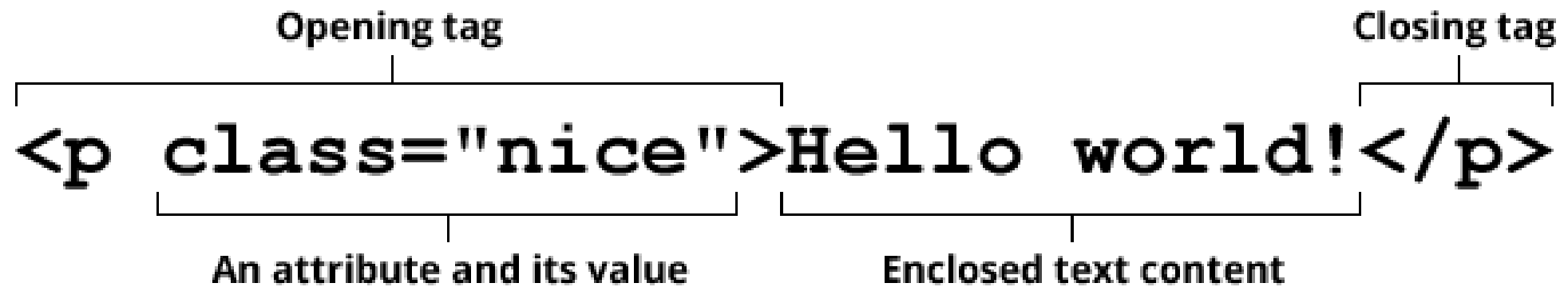
Inside the <head> Element

- Title: <title> element for webpage title.
- Metadata: <meta> elements for character encoding, description, and keywords.
- Viewport: <meta name="viewport"> for responsive design settings.
- Stylesheets: <link> or <style> for CSS.
- JavaScript: <script> for interactivity.
- Favicon: <link rel="icon"> for favicon.
- External Resources: Any additional external resources.

Elements, Tags, and Attributes



Anatomy of an HTML element



Text Formatting

- Headings:

Use `<h1>` to `<h6>` for headings of varying importance.

- Paragraphs:

Use `<p>` for organizing text into paragraphs.

- Bold and Italic:

`<strong>` for bold text.

`<em>` for italicized text.

Links

- Creating Hyperlinks: Use `<a>` element to create hyperlinks.

```
<a href="URL">Link Text</a>
```

- Linking to External URLs: Specify the URL in the href attribute to link to external websites.

```
<a href="https://example.com">External Link</a>
```

- Linking to Internal Pages: Use relative URLs to link to other pages within the same website.

```
<a href="page.html">Internal Link</a>
```

Lists

- Ordered Lists:

- Use `<ol>` for ordered lists.
- Use `<li>` for list items.

```
<ol>
  <li>Item 1</li>
  <li>Item 2</li>
  <li>Item 3</li>
</ol>
```

- Unordered Lists:

- Use `<ul>` for unordered lists.
- Use `<li>` for list items.

```
<ul>
  <li>Item 1</li>
  <li>Item 2</li>
  <li>Item 3</li>
</ul>
```

Description Lists (dl)

- Description Lists:
- Use <dl> for description lists.
- Use <dt> for the term (name)
- <dd> for the description (value).

```
<dl>
  <dt>Term 1</dt>
  <dd>Description 1</dd>
  <dt>Term 2</dt>
  <dd>Description 2</dd>
  <dt>Term 3</dt>
  <dd>Description 3</dd>
</dl>
```

Images

- Inserting Images:

Use `<img>` element to insert images.

- Specifying Image Attributes:

- `src`: Specifies the image source (URL or file path).
- `alt`: Provides alternative text for accessibility and SEO.
- `width` and `height`: Specify image dimensions (optional).

Audio

- Embedding Audio:

Use <audio> element to embed audio files.

- Specifying Audio Source:

Use the “src” attribute to specify the audio file source (URL or file path).

```
<audio src="audio.mp3" controls></audio>
```

- Additional attributes: ‘controls’, ‘autoplay’, ‘loop’, ‘muted’

Video

- Embedding Video:

Use <video> element to embed audio files.

- Specifying Video Source:

Use the “src” attribute to specify the audio file source (URL or file path).

```
<video src="video.mp4" controls></video>
```

- Additional attributes: ‘controls’, ‘autoplay’, ‘loop’

Tables

- Creating Tables:
 - Use `<table>` for creating tables.
 - Use `<tr>` for table rows
 - `<th>` for table headers
 - `<td>` for table cells.

```
<table>
  <tr>
    <th>Header 1</th>
    <th>Header 2</th>
  </tr>
  <tr>
    <td>Row 1, Cell 1</td>
    <td>Row 1, Cell 2</td>
  </tr>
  <tr>
    <td>Row 2, Cell 1</td>
    <td>Row 2, Cell 2</td>
  </tr>
</table>
```

iframes

- Embedding External Content:
- Use `<iframe>` element to embed external content within a webpage.
- Specifying Source:
- Use the `src` attribute to specify the source URL of the content to be embedded.
- Additional attributes: `'height'`, `'width'`, `'frameborder'`, `'scrolling'`